

FAR WESTERN UNIVERSITY
Semester End Examination-2081

English Grammar and Composition (CSIT.111)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: First

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What types of food do supertasters avoid?
2. Mention the things that most people give up or sacrifice in order to be successful.
3. What causes major injuries in child athletes?
4. Would you consider escaping from television for a month or more? Why or why not?
5. Do you think bystander effect is effective in every situation? Why or why not?
6. Most people believe that advertising should not be taken seriously. Do you agree?
7. What does 'take a risk' mean? Explain.
8. How would you prepare yourself to do an extreme sport?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. How has Second Life become more like the real world? Explain with examples given in the text. Give your opinion on the popularity of this website.
10. What are mirror neurons? How do they help us assimilate with the emotions of other people?

11. 'Urban Farming' can be taken as a symbolic idea to build greener cities. How can this approach be made effective. Give your suggestions.
12. Differentiate between nontaster, medium taster and super taster. How can nontasters and supertasters be affected by their choices of food? Discuss.
13. Do you think life in your city/town is becoming difficult because of increasing pollution? What can you and other people do to deal with this problem?
14. What qualities do you think Beatrice has that have helped her to be successful in her goals?

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Discuss the advantages and disadvantages of living in a city like Masdar?
16. "People are happier from experiences than from possessions". Do you agree or not? Give reasons in support of your view.
17. What kind of topics should be avoided when one meets someone for the first time? Why do you think such topics need to be avoided?
18. How do you perceive the effect of social media in communication? Give suitable examples.

FAR WESTERN UNIVERSITY
Semester End Examination-2081

Information Technology Fundamentals (CSIT.112)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Explain ASCII code.
2. Differentiate between static and dynamic RAMs.
3. Convert $(2B.DB)_{16}$ into octal number system.
4. Define data files.
5. Define database. What are the benefits of using databases?
6. What is cache memory? How it is different from register?
7. Why do you use search engines?
8. State and prove De-Morgan's theorem.

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. What are the main features and examples of fourth generation computers?
10. Define decimal number system. Subtract $(567)_{10}$ from $(68)_{10}$ by using 9's and 10's complement method.
11. Explain half adder and full adder circuit with logical expression and truth table.
12. Explain classification of computers on the basis of size with examples.
13. What is programming language? Describe the various types of programming languages.
14. Write short notes on: (any two)
 - a. Modem
 - b. Non-impact printer
 - c. Internet service provider

P.T.O.

School/Campus:.....

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

- 15) Classify the operating system into different types based on their processing capability. List the main function of the operating system and describe in details.
- 16) What is Boolean algebra? Describe any five logic gates with truth tables and logical symbols.
- 17) Compare system software with application software. Explain different types of software in brief.
- 18) Explain the concept of computer networks. Discuss the importance of networking in computer systems. Differentiate between LAN and WAN network.

FAR WESTERN UNIVERSITY

Semester End Examination-2081

Calculus and Analytical Geometry (CSIT.113)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Show that the series $\sum_{n=0}^{\infty} \frac{2(-1)^n}{5 \times 3^n}$ converges to $\frac{3}{10}$
2. Find the equation of the parabola whose vertex is at (-1,1) and whose directrix is $y = 3$
3. Find the angle between the vectors
 $\vec{u} = 4\vec{i} - 2\vec{j} - \vec{k}$ and $\vec{v} = 4\vec{i} - 2\vec{j} + 4\vec{k}$
4. Find volume of the parallelepiped determined by vectors
 $\vec{u} = \vec{i} + \vec{j} - \vec{k}$, $\vec{v} = -2\vec{i} + 3\vec{k}$ and $\vec{w} = 7\vec{j} - 4\vec{k}$
5. Find the n^{th} derivative of $\frac{1}{x^2 - a^2}$
6. Find f_x , f_y and f_{xy} of the function $f(x,y) = 3x^2 - 4xy + 2y^2$
7. Show that $\Gamma \frac{9}{2} = \frac{105}{16} \pi$
8. Find the reduction formula of $\int \tan^n x dx$

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. State ratio test theorem. Use it to test the convergence of the series
 $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$
10. Find the co-ordinates of center, vertices, eccentricity, foci, length of transverse and conjugate axis of the hyperbola $5x^2 - 4y^2 - 20x - 8y = 4$
11. Find the distance from the point S to the line L which passes through P_0 and parallel to $\vec{v}$ and hence find the distance from the point (1, 3, 2) to the line $x = 2 + 2t$, $y = 1 + 6t$, $z = 3$

P.T.O.

12. i. If $u = \log \sqrt{x^2 + y^2 + z^2}$ show that $u_{xx} + u_{yy} + u_{zz} = \frac{1}{x^2 + y^2 + z^2}$

ii. If $V = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ show that $V_{xx} + V_{yy} + V_{zz} = 0$

13. Find the reduction formula of $\int \cos^m x \sin^n x dx$

14. Integrate the followings

i. $\int_{\pi}^{2\pi} \int_0^{\pi} (\sin x + \cos y) dx dy$

ii. $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$

Group - 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. a. Give the statement of the integral test. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ by integral test.

b. Give the statement of Cauchy n^{th} root test. Examine the convergence of the series $\sum \left(1 + \frac{1}{\sqrt{n}}\right)^{-n^{\frac{3}{2}}}$

16. Apply gamma function prove the followings

i. $\int_0^{\pi/2} \sin^4 x \cos^2 x dx = \frac{\pi}{32}$

ii. $\int_0^1 x^6 \sqrt{1-x^2} dx = \frac{5\pi}{256}$

iii. Use property of definite integral prove that $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{1+\sqrt{\cot x}} dx = \frac{\pi}{4}$

17. If $\vec{v} = 2\vec{i} + 10\vec{j} - 11\vec{k}$ and $\vec{u} = 2\vec{i} + 2\vec{j} + \vec{k}$ find the followings

i. $\vec{u} \cdot \vec{v}$

ii. $\vec{u} \times \vec{v}$

iii. the cosine of angle between $\vec{u}$ and $\vec{v}$

iv. the vector projection of $\vec{u}$ on $\vec{v}$

18. Give the statement of the Leibnitz's theorem for n^{th} derivative of product of two functions. If $y = e^{a \tan^{-1} x}$, prove that

i. $(1+x^2)y_2 + (2x-a)y_1 = 0$

ii. $(1+x^2)y_{n+2} + (2nx+2x-a)y_{n+1} + n(n+1)y_n = 0$

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Electronic Principles (CSIT.114)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. State Kirchoff's first and second law.
2. What is the difference between photodiode and LED?
3. What are alpha (α) and beta (β) of a transistor? Write the relation between α and β .
4. What do you mean by JFET? How is it different from ordinary transistor?
5. What do you mean by virtual ground in operational amplifier? Explain it by drawing figure.
6. State the condition under which feedback amplifier works as oscillator?
7. Distinguish between regulated and unregulated power supply.
8. What is current booster?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

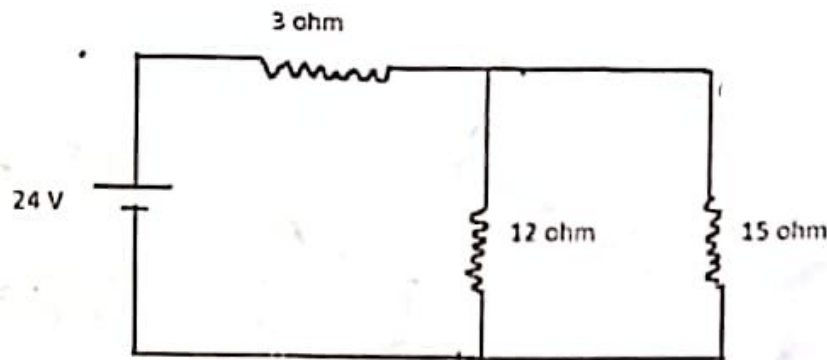
9. Distinguish between avalanche and zener effect? How the Zener diode can be used as voltage regulator?
10. How transistor can be used as an amplifier and as a switch.
11. Draw the circuit diagram of common base NPN transistor. Also discuss input, output and current transfer characteristics of common base NPN transistor.
12. Define operational amplifier? Show that how operational amplifier can be used as adder and subtractor.
13. Draw the circuit diagram of Colpitts oscillator and explain its working.
14. Write short notes on following topics:
 - (a) Intrinsic semiconductor
 - (b) Extrinsic semiconductor
 - (c) Series voltage regulator
 - (d) Shunt voltage regulator

Group – 'C'

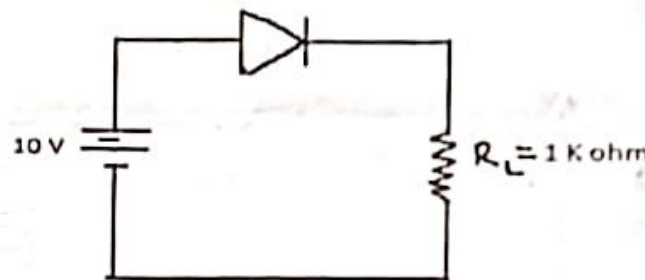
Attempt any three questions. (Long answer questions)

3×12 = 36

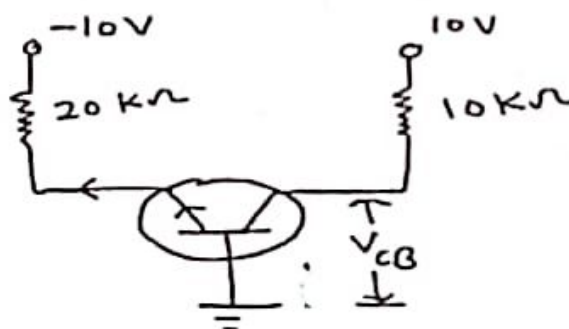
15. (a) State and explain the Norton's theorem by taking an example of arbitrary circuit diagram. [8]
 (b) Find the current through 15 ohm resistor using Thevenin's theorem. [4]



16. (a) What is rectifier? Describe the full wave rectifier. [8]
 (b) In the circuit given below, find load voltage, load current and diode power. (Given barrier potential of diode is 0.7 V).



17. (a) Discuss the JFET drain characteristics when $V_{GS} = 0$. [8]
 (b) For a certain JFET, $I_{DSS} = 16 \text{ mA}$ and $V_{GS(off)} = -5 \text{ V}$. Determine drain current when $V_{GS} = -4 \text{ V}$ [4]
 18. (a) Draw the circuit diagram of common emitter amplifier and discuss its working? Why there is phase inversion between input and output signal. [8]
 (b) For the circuit given below draw the dc load line. [4]



FAR WESTERN UNIVERSITY
Semester End Examination-2081

Programming Fundamentals and C Programming (CSIT.115)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Define control structure. What are the various decision making statements available in C?
2. What is Pointer? What are the uses of Pointers?
3. Name different types of errors which can occur during the execution of a program? When a syntax error occurs?
4. Write down the difference between compiler and interpreter.
5. Write a program that inputs radius from the user and calculates area and circumference of a circle.
6. Write the features of c programming language.
7. Define dangling pointer variable.
8. What do you mean by the scope of the variable?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. What do you mean by algorithm and flowchart? Explain with examples.
10. Differentiate between while and do-while loop. Write C program to perform various arithmetic operation on two operands.
11. What is operator? Describe different types of operators in C programming with suitable examples.
12. Describe the working of for loop and while loop with flowchart and examples.
13. Write C program to enter numbers until user press 0 and display sum and average of entered number.
14. Write C program to store n numbers in a list and enter a number. Check whether the entered number is found in list or not.

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Describe string. Explain different string manipulation function with examples.
16. Define structure. Explain nested structure with example. Create a structure named **book** with **name**, **author**, and **publisher** as its members. Write a program using this structure to read data of 50 books and display name of those books published by "XYX" publisher.
17. Explain components of functions. Write C program to display factorial of entered number by using recursion.
18. What are file opening modes? Explain them. Write a program to open a new file, read name, address and marks of n students from user and write to a file.
